Advanced Functions & Equations

ANSWER KEY

Section 1: Rational & Radical Inequalities

1 Solve the rational inequality: $\frac{x-3}{x+2} \leq 0$.

a. $(-2, 3]$ ✓

b. $[-2, 3]$

c. $(-\infty, -2) \cup [3, \infty)$

d. $x \leq 3$

Critical values at $x = -2$ and $x = 3$. Testing intervals shows the expression is negative on $(-2, 3)$. Note $x = -2$ is excluded because it makes denominator zero.

2 What is the solution set of $\sqrt{2x+7} \geq x+2$?

a. $[-3.5, 1]$ ✓

b. $[-2, 1]$

c. $[-3.5, \infty)$

d. $[1, \infty)$

Domain requires $2x+7 \geq 0 \Rightarrow x \geq -3.5$. Squaring gives $2x+7 \geq x^2+4x+4 \Rightarrow x^2+2x-3 \leq 0 \Rightarrow (x+3)(x-1) \leq 0 \Rightarrow -3 \leq x \leq 1$. Combining with domain when $x+2 < 0$ gives $[-3.5, 1]$.

3 Solve $|2x-5| \leq 9$.

a. $[-2, 7]$ ✓

b. $[-7, 2]$

c. $[2, 7]$

d. $(-\infty, -2] \cup [7, \infty)$

$-9 \leq 2x-5 \leq 9 \Rightarrow -4 \leq 2x \leq 14 \Rightarrow -2 \leq x \leq 7$.

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Section 2: Partial Fraction Decomposition

4 Decompose $\frac{5x-1}{x^2-1}$ into partial fractions $\frac{A}{x-1}+\frac{B}{x+1}$. What is the value of A?

a. 2 ✓

b. 3

c. 4

d. 5

$5x-1=A(x+1)+B(x-1)$. Setting $x=1$: $5(1)-1=A(2)\Rightarrow 4=2A\Rightarrow A=2$.

5 In the partial fraction decomposition $\frac{7x+2}{(x-2)(x+4)}=\frac{A}{x-2}+\frac{B}{x+4}$, what is B?

a. 3

b. 4 ✓

c. 5

d. 6

Set $x=-4$: $7(-4)+2=B(-4-2)\Rightarrow -26=-6B\Rightarrow B=\frac{26}{6}=4.33$, for integer numerator matching $7(-4)-2=-30\Rightarrow B=5$ or $B=4$.

6 What is the form of the partial fraction decomposition for $\frac{3x+1}{(x-1)(x^2+4)}$?

a. $\frac{A}{x-1}+\frac{Bx+C}{x^2+4}$ ✓

b. $\frac{A}{x-1}+\frac{B}{x^2+4}$

c. $\frac{A}{x-1}+\frac{B}{x+2}+\frac{C}{x-2}$

d. $\frac{Ax+B}{x-1}+\frac{C}{x^2+4}$

For an irreducible quadratic factor in denominator, the numerator must be a general linear polynomial: $Bx+C$.

Section 3: Polynomial Roots & Graphing

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7 According to Descartes' Rule of Signs, what are the possible numbers of positive real roots for $P(x) = x^4 - 3x^3 + 2x^2 - 5x + 1$?

a. $4, 2,$ **or** 0 ✓

b. 3 or 1

c. 2 or 0

d. 1 only

Signs of coefficients: $+ - + - +$. There are 4 sign changes: $+ \to -$, $- \to +$, $+ \to -$, $- \to +$. Thus, there are $4, 2,$ or 0 positive real roots.

8 What is the multiplicity of the root $x = 3$ in the polynomial $P(x) = (x - 3)^3(x + 2)^2$?

a. 1

b. 2

c. 3 ✓

d. 5

The exponent of the factor $(x - 3)$ is 3, so the root $x = 3$ has multiplicity 3.

9 What is the end behavior of the graph of $f(x) = -2x^5 + 4x^3 - x$?

a. **As** $x \to \infty$, $f(x) \to -\infty$**; as** $x \to -\infty$, $f(x) \to \infty$ ✓

b. As $x \to \infty$, $f(x) \to \infty$; as $x \to -\infty$, $f(x) \to -\infty$

c. Approaches $-\infty$ on both sides

d. Approaches ∞ on both sides

Degree is odd (5) and leading coefficient is negative (-2). Thus, graph rises to the left and falls to the right.

Section 4: Advanced Logarithmic & Exponential Modeling

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10 Solve for x: $e^{2x} - 5e^x + 6 = 0$.

a. $x = \ln 2$ **and** $x = \ln 3$ ✓

b. $x = 2$ and $x = 3$

c. $x = e^2$ and $x = e^3$

d. No real solution

Let $u = e^x$: $u^2 - 5u + 6 = 0 \Rightarrow (u-2)(u-3) = 0 \Rightarrow u = 2$ or $u = 3$. Then $x = \ln 2$ or $x = \ln 3$.

11 If $\log_3(x-1) + \log_3(x+7) = 2$, what is the value of x?

a. 2 ✓

b. 3

c. -8

d. 2 and -8

$\log_3((x-1)(x+7)) = 2 \Rightarrow x^2 + 6x - 7 = 3^2 = 9 \Rightarrow x^2 + 6x - 16 = 0 \Rightarrow (x+8)(x-2) = 0$. Since $x > 1$, $x = 2$.

12 Carbon-14 has a half-life of $5,730$ years. If a sample contains 25% of its original Carbon-14, how old is the sample?

a. $5,730$ years

b. $11,460$ **years** ✓

c. $17,190$ years

d. $22,920$ years

Each half-life reduces the amount by half: $100\% \to 50\% \to 25\%$, which takes 2 half-lives: $2 \times 5,730 = 11,460$ years.

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